1. What is Responsive UI Design in Flutter?

Responsive UI design means designing an application that automatically adjusts its layout according to different screen sizes, such as mobile phones and tablets.

2. What is MediaQuery in Flutter?

MediaQuery is used to obtain device screen information, such as screen width, height and orientation.

Example:

```
double width = MediaQuery.of(context).size.width;
```

3. What are Breakpoints in responsive design?

Breakpoints are predefined screen-width values used to change the layout according to the device size.

Example:

```
if (width < 600) {
  // Mobile layout
} else {
  // Tablet layout
}
```

4.What is the difference between MediaQuery and LayoutBuilder?

MediaQuery provides information about the device screen.

LayoutBuilder provides the available space and constraints from its parent widget.

5. What is OrientationBuilder?

OrientationBuilder is a widget that helps build different layouts depending on whether the device is in portrait or landscape orientation.

6. What is Navigator in Flutter?

Navigator is a widget that manages the navigation stack and helps move between different screens in an application.

7. What is Navigator.push()?

Navigator.push() is used to open a new screen by adding a new route to the navigation stack.

Example:

```
Navigator.push(
  context,
  MaterialPageRoute(
    builder: (context) => ProductDetails(),
  ),
);
```

8. What is Navigator.pop()?

Navigator.pop() is used to close the current screen and return to the previous screen.

Example:

```
Navigator.pop(context);
```

9. Explain the navigation flow in an Online Shopping App.

The navigation flow is:

Home → Product Details → Cart.

- Home screen displays available products.
- Product Details screen displays the selected product information.
- Cart screen displays the selected products.

10. What is MaterialPageRoute?

MaterialPageRoute is used to create a route between screens with Material Design transition effects.

11. What is a Custom Widget in Flutter?

A custom widget is a user-defined widget created to reuse UI components in different parts of an application.

Example: ProductCard, ProfileCard and CustomButton.

12. What are the two main types of widgets in Flutter?

StatelessWidget

StatefulWidget

13. What is the difference between StatelessWidget and StatefulWidget?

StatelessWidget	**StatefulWidget**
UI does not change dynamically due to internal state changes.	UI can change dynamically.
Does not maintain mutable state.	Maintains state using a separate State object.
Example: Static text.	Example: Counter application.

14. What is ThemeData in Flutter?

ThemeData is used to define the common appearance of an application, such as colours, fonts and button styles.

Example:

```
MaterialApp(
  theme: ThemeData(
    primaryColor: Colors.blue,
  ),
```

```
  home: HomePage(),
);
```

15. What is the difference between Padding and Margin?

- Padding is the space inside a widget's boundary.
- Margin is the space outside a widget's boundary.

16. What is a Form widget in Flutter?

Form is a widget used to group multiple input fields and manage their validation.

17. What is TextFormField?

TextFormField is used to accept user input and supports validation through its validator property.

18. What is GlobalKey in Flutter Forms?

GlobalKey uniquely identifies a widget and allows us to access its state.

Example:

```
final _formKey = GlobalKey<FormState>();
```

19. How do you validate a form in Flutter?

We use the validate() method of FormState.

Example:

```
if (_formKey.currentState!.validate()) {
  print("Form is valid");
}
```

20. What is the purpose of the validator property?

The validator property checks the entered data and returns an error message if the input is invalid. It returns null if the input is valid.

Example:

```
validator: (value) {
  if (value == null || value.isEmpty) {
    return 'Enter your name';
  }
  return null;
}
```

21. What is Animation in Flutter?

Animation is the process of creating movement or visual changes in UI elements over a specified time.

22. What is FadeTransition?

FadeTransition gradually changes the opacity of a widget, making it appear or disappear.

23. What is SlideTransition?

SlideTransition moves a widget from one position to another during an animation.

24. What is AnimationController?

AnimationController controls the progress, duration and direction of an animation.

25. What is the difference between FadeTransition and SlideTransition?

FadeTransition	**SlideTransition**
Changes opacity.	Changes position.
Creates fade-in and fade-out effects.	Creates sliding effects.
Controls visibility.	Controls movement.